

ArcVoucher: A Unified Balance-Based Digital Gift Card Marketplace for Stablecoin-Native Commerce on Arc Network

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Abstract

The rapid adoption of stablecoins has transformed digital payments and decentralized finance. However, blockchain-based commerce remains fragmented due to network-specific balances, bridge requirements, and inconsistent user experiences. This research introduces ArcVoucher, a stablecoin-native digital gift card marketplace built on Arc Network and Arc Unified Balance. ArcVoucher enables users to purchase digital vouchers using USDC through either direct Arc payments or Unified Balance aggregation across multiple supported blockchains. The system integrates smart contracts, a fulfillment backend, wallet-bound voucher ownership, and cross-chain payment abstraction. A quantitative evaluation framework is proposed to measure transaction efficiency, fulfillment success rate, and user interaction complexity. Experimental implementation on Arc Testnet demonstrates that Unified Balance significantly simplifies digital commerce workflows while preserving blockchain transparency, ownership verification, and payment security.

Keywords: Arc Network, Unified Balance, Stablecoin Commerce, Digital Voucher Marketplace, Cross-Chain Payments, USDC, Blockchain Applications.

I. INTRODUCTION

A. Background

Stablecoins have emerged as the dominant payment medium within blockchain ecosystems due to their low volatility and high interoperability. According to Circle and Arc Network documentation, stablecoins are expected to become foundational infrastructure for internet-native commerce.

Despite their advantages, users still encounter significant barriers:

1. Asset fragmentation across chains.

2. Manual bridge operations.
3. Repeated network switching.
4. Complex payment settlement processes.

These challenges reduce the usability of blockchain applications for mainstream consumers.

Arc Network introduces Unified Balance, a chain abstraction mechanism that aggregates USDC across supported blockchains into a single spendable balance. This capability enables new forms of digital commerce that resemble

traditional fintech experiences while maintaining blockchain settlement guarantees.

B. Problem Statement

Current blockchain commerce systems suffer from:

- Multi-chain balance fragmentation.
- High user interaction complexity.
- Bridge dependency.
- Poor payment experience.

Consequently, there is a need for a commerce platform that abstracts cross-chain complexity while preserving user ownership and payment transparency.

C. Research Objectives

This research aims to:

1. Design a stablecoin-native digital voucher marketplace.
 2. Integrate Arc Unified Balance into a real commerce workflow.
 3. Measure payment efficiency and fulfillment success.
 4. Evaluate Unified Balance as a commerce primitive.
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D. Research Contributions

This paper contributes:

1. A practical implementation of Arc Unified Balance.
2. A digital commerce architecture utilizing stablecoin aggregation.

3. A quantitative evaluation model for blockchain commerce systems.
 4. A reference implementation for future Arc ecosystem applications.
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II. LITERATURE REVIEW

A. Stablecoin-Based Commerce

Stablecoins provide predictable value representation and reduce volatility risk compared to native crypto assets.

Several studies identify stablecoins as a key enabler for:

- Digital payments
 - E-commerce
 - Cross-border transactions
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B. Chain Abstraction

Chain abstraction seeks to eliminate blockchain-specific complexity from users.

Traditional Model:

User → Bridge → Destination Chain → Application

Chain-Abstraction Model:

User → Unified Balance → Application

Arc Unified Balance represents a practical implementation of this concept.

C. Digital Voucher Marketplaces

Digital gift cards represent a widely adopted digital asset category.

Common requirements include:

- Secure ownership verification
- Inventory management
- Instant fulfillment
- Fraud prevention

Blockchain technology provides verifiable ownership and transparent settlement mechanisms.

III. RESEARCH METHODOLOGY

A. Research Design

This study uses a quantitative experimental approach.

Independent variables influence transaction performance and user experience metrics.

B. Research Hypotheses

H1

Unified Balance reduces transaction complexity compared to direct multi-chain payment workflows.

H2

Unified Balance increases payment completion efficiency.

H3

Wallet-bound voucher ownership improves fulfillment security.

C. Variables

Independent Variables

Cod e	Variable	Description
X1	Unified Balance Usage	Use of Unified Balance during checkout
X2	Number of Source Chains	Chains contributing USDC
X3	Available Unified Balance	Total spendable USDC

Dependent Variables

Code	Variable	Description
Y1	Completion Time	Time from checkout to confirmation
Y2	Fulfillment Success Rate	Successful voucher issuance
Y3	Interaction Complexity	Number of user actions

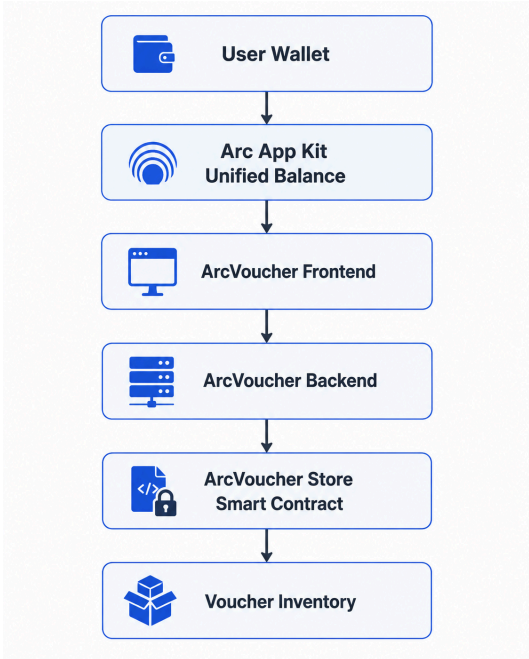
D. Operational Variable Table

Variable	Indicator	Unit
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Unified Balance Usage	Enabled / Disabled	Binary
Source Chains	Count	Integer
Balance	USDC Amount	USDC
Completion Time	Seconds	Seconds
Fulfillment Success	Percentage	%
Complexity	User Actions	Count

IV. SYSTEM ARCHITECTURE

A. Architecture Overview



B. Frontend Layer

Technologies:

- Next.js
- TypeScript
- Arc App Kit
- Viem

Functions:

- Product browsing
- Checkout
- Order history
- Voucher reveal

C. Smart Contract Layer

ArcVoucherStore

Functions:

- Product inventory
- Order management
- Fulfillment verification

Key methods:

- `purchase()`
- `addStock()`
- `fulfillOrder()`

D. Backend Layer

Functions:

- Voucher generation
- Order indexing
- Unified Balance verification
- Voucher persistence

Deployment:

- Railway

V. MATHEMATICAL MODEL

Unified Balance Aggregation

Let:

B_i = Balance on chain i

n = Number of chains

Unified Balance:

$$UB = \sum B_i$$

for $i = 1$ to n

Purchase Validity

Purchase is accepted if:

$$UB \geq P + F$$

Where:

P = Product Price

F = Estimated Transaction Fee

Fulfillment Success Rate

$$FSR = (S / T) \times 100\%$$

Where:

S = Successful Fulfillments

T = Total Orders

VI. IMPLEMENTATION

Environment

Frontend:

- Next.js
- Vercel

Backend:

- Node.js
- Railway

Blockchain:

- Arc Testnet

Stablecoin:

- USDC
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Supported Products

1. Steam Gift Card
 2. Epic Games Gift Card
 3. Amazon Gift Card
 4. Apple Gift Card
 5. Google Play Gift Card
 6. Netflix Gift Card
 7. Spotify Gift Card
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Payment Modes

1. Direct Arc Payment

Wallet → ArcVoucherStore

2. Unified Balance Payment

Wallet → Unified Balance → Arc Gateway
→ ArcVoucher

VII. RESULTS AND DISCUSSION

Experimental Results

The implementation successfully demonstrated:

- Direct payment settlement

- Unified Balance deposit
- Unified Balance spending
- Voucher fulfillment
- Wallet-bound voucher ownership

User Complexity Reduction

Traditional workflow:

1. Connect Wallet
2. Switch Chain
3. Bridge Funds
4. Wait Settlement
5. Switch Chain
6. Pay

Unified Balance workflow:

1. Connect Wallet
2. Deposit USDC
3. Pay

This significantly reduces interaction complexity.

Security Analysis

ArcVoucher maintains security through:

- Wallet ownership verification
 - Order-based voucher authorization
 - On-chain payment verification
 - Voucher hash tracking
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Observed Challenges

1. Decimal normalization (6 vs 18 decimals).
2. Backend persistence.
3. Unified Balance verification edge cases.

All issues were resolved through normalization and persistent storage strategies.

VIII. CONCLUSION

ArcVoucher demonstrates the viability of stablecoin-native digital commerce built on Arc Unified Balance.

The research confirms that:

1. Unified Balance simplifies cross-chain commerce.
2. Stablecoin aggregation improves payment usability.
3. Wallet-bound voucher systems provide secure digital delivery.
4. Arc infrastructure can support real-world marketplace applications.

Future work includes PostgreSQL integration, merchant dashboards, real voucher inventory management, and advanced analytics

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